

# Comparison of Deep Learning Architectures

## **Submitted by:**

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Course: BCA – AI/ML

Subject: Deep Learning

Date:31/08/2026

## **❖ Introduction**

Explain what deep learning is and why different architectures are used.

### **Introduction**

Deep Learning is a branch of Artificial Intelligence that uses artificial neural networks with multiple layers to learn patterns from data. Different deep learning architectures are designed for different types of data and problems.

Three commonly used architectures are:

1. Feedforward Neural Networks (FNN)
2. Convolutional Neural Networks (CNN)
3. Recurrent Neural Networks (RNN)

Each architecture has a different structure, works with different types of input data, and is suitable for different real-world applications.

## **❖ Objectives**

### **Objectives of the Project**

- To understand the basic architecture of Feedforward Neural Networks.
- To understand the basic architecture of Convolutional Neural Networks.
- To understand the basic architecture of Recurrent Neural Networks.
- To identify applications of each architecture.
- To compare FNN, CNN, and RNN based on structure and input data.
- To understand which architecture is suitable for different problems.

## ❖ Task 1 – Study the Basic Architectures

### ➤ Feedforward Neural Network (FNN)

A Feedforward Neural Network is one of the simplest neural network architectures. Information moves only in one direction—from the input layer to the hidden layers and finally to the output layer.

Basic Structure

Input Layer → Hidden Layer(s) → Output Layer

Example:

Input Layer



Hidden Layer



Hidden Layer



Output Layer

### Characteristics

- Information flows in one direction.
- Does not contain feedback connections.
- Suitable for structured/tabular data.
- Relatively simple architecture.
- Can be used for classification and regression.

### Applications

- Student performance prediction
- Sales prediction
- Customer classification
- Credit risk prediction
- House price prediction

## ➤ Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is a deep learning architecture mainly designed for processing images and visual data.

CNNs use convolutional layers to identify features such as edges, shapes, textures, and objects.

Basic Structure

Input Image



Convolution Layer



Pooling Layer



Convolution Layer



Pooling Layer



Fully Connected Layer



Output

## Characteristics

- Designed mainly for spatial data.
- Automatically extracts important features.
- Uses convolution and pooling operations.
- Very effective for image recognition.
- Can handle large visual datasets.

## Applications

- Face recognition

- Image classification
- Object detection
- Medical image analysis
- Self-driving vehicles
- Facial expression recognition

### ➤ **Recurrent Neural Network (RNN)**

A Recurrent Neural Network (RNN) is designed to process sequential or time-dependent data.

Unlike FNNs, RNNs can use information from previous steps to help process the current step.

Basic Structure

Input 1 → RNN → Output 1

↓

Input 2 → RNN → Output 2

↓

Input 3 → RNN → Output 3

The network maintains information from previous inputs through its hidden state.

### **Characteristics**

- Designed for sequential data.
- Maintains information from previous inputs.
- Useful for time-series data.
- Can process sequences of different lengths.
- Basic RNNs can face vanishing-gradient problems for long sequences.

### **Applications**

- Speech recognition
- Language modelling

- Text generation
- Sentiment analysis
- Stock/time-series prediction
- Machine translation

### ❖ Task 2 – Applications

| Architecture | Suitable Data               | Common Applications                    |
|--------------|-----------------------------|----------------------------------------|
| <b>FNN</b>   | Tabular/structured data     | Prediction, classification, regression |
| <b>CNN</b>   | Images/visual data          | Face recognition, object detection     |
| <b>RNN</b>   | Sequential/time-series data | Speech, text, forecasting              |

### ❖ Task 3 – Comparison

| Feature            | FNN                                    | CNN                                            | RNN                              |
|--------------------|----------------------------------------|------------------------------------------------|----------------------------------|
| Full Name          | Feedforward Neural Network             | Convolutional Neural Network                   | Recurrent Neural Network         |
| Main Data          | Structured data                        | Images/spatial data                            | Sequential data                  |
| Information Flow   | One direction                          | Forward through layers                         | Uses previous information        |
| Main Layers        | Input, hidden, output                  | Convolution, pooling, fully connected          | Input, hidden, recurrent, output |
| Memory             | No memory                              | No traditional sequence memory                 | Maintains hidden state           |
| Feature Extraction | Requires learned dense representations | Excellent automatic spatial feature extraction | Learns patterns across sequences |
| Best Used For      | Classification & regression            | Image and vision tasks                         | Time-series and language tasks   |

|         |                  |                  |                    |
|---------|------------------|------------------|--------------------|
| Example | Sales prediction | Face recognition | Speech recognition |
|---------|------------------|------------------|--------------------|

## Simple Real-World Example

This will make your project better.

Suppose we want to solve three different problems:

Problem 1: Predict house prices

Use FNN.

House Size

Number of Rooms

Location

Age

↓

FNN

↓

Predicted House Price

Problem 2: Identify a cat in an image

Use CNN.

Image

↓

CNN

↓

Feature Extraction

↓

Cat / Not Cat

Problem 3: Predict the next word in a sentence

Use RNN.

"I am going to..."

↓

RNN

↓

"college"

This shows why different architectures are used for different types of data.

## ❖ Advantages and Limitations

### FNN

#### Advantages

- Simple architecture.
- Easy to understand.
- Good for structured data.
- Useful for classification and regression.

#### Limitations

- Does not remember previous inputs.
- Not ideal for images or sequences.
- May require feature engineering depending on the problem.

### CNN

#### Advantages

- Excellent for image processing.
- Automatically extracts visual features.
- Good at recognizing spatial patterns.

#### Limitations

- Usually requires substantial training data.
- Computationally expensive for large models.
- Mainly designed around spatial structure.

# **RNN**

## **Advantages**

- Suitable for sequential data.
- Can use information from previous time steps.
- Useful for text and time-series problems.

## **Limitations**

- Training can be difficult for long sequences.
- Basic RNNs can suffer from vanishing gradients.
- Sequential processing can be slower than highly parallel architectures.

## **❖ Conclusion**

Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks are important deep learning architectures designed for different types of problems.

FNNs are suitable for structured data and basic prediction or classification tasks. CNNs are highly effective for images and other spatial data because they can automatically learn visual features. RNNs are designed for sequential and time-dependent data because they can use information from previous steps.

Therefore, the choice of architecture depends mainly on the type of input data and the problem being solved.

## **❖ Learning Outcomes**

After completing this project, students will be able to:

- Explain the basic structure of FNN, CNN, and RNN.
- Understand the differences between the three architectures.
- Identify suitable applications for each architecture.
- Select an appropriate architecture based on the type of data.
- Understand the importance of deep learning architectures in real-world applications.



## ❖ References

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville – *Deep Learning*.
2. Michael Nielsen – *Neural Networks and Deep Learning*.
3. Deep Learning course materials and classroom notes.
4. Research articles and educational resources on neural networks.